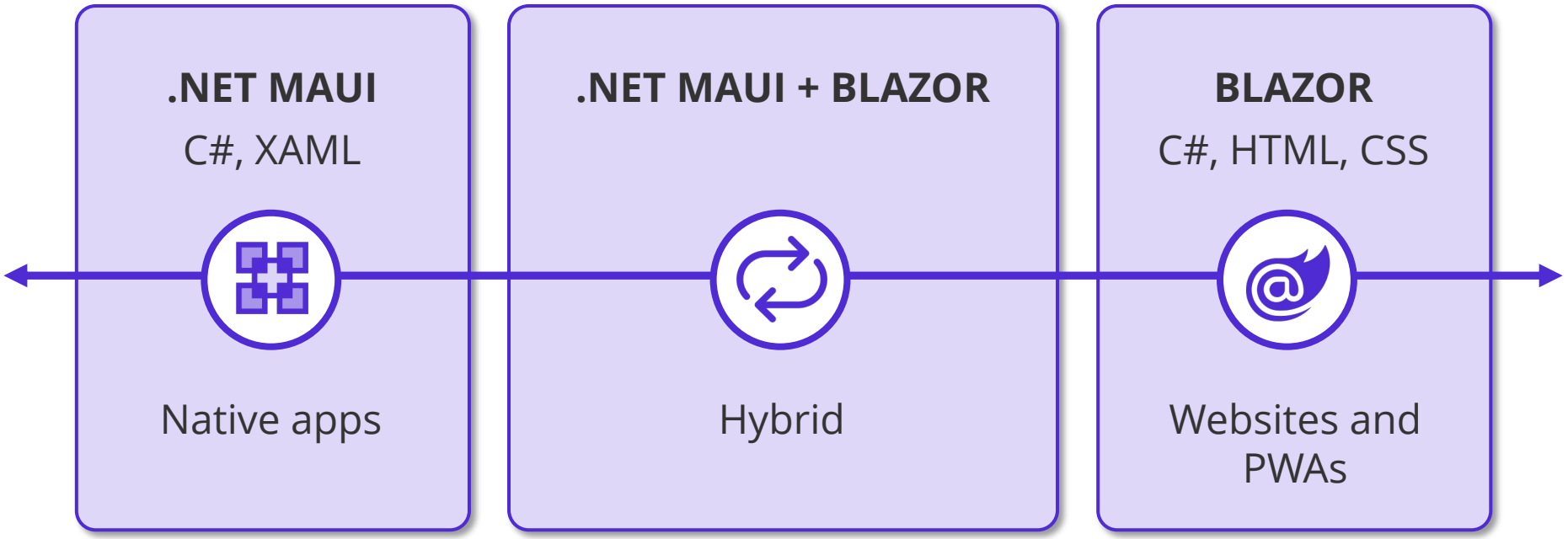
Build Hybrid Apps with .NET MAUI & Blazor

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.NET offers choice for client app developers

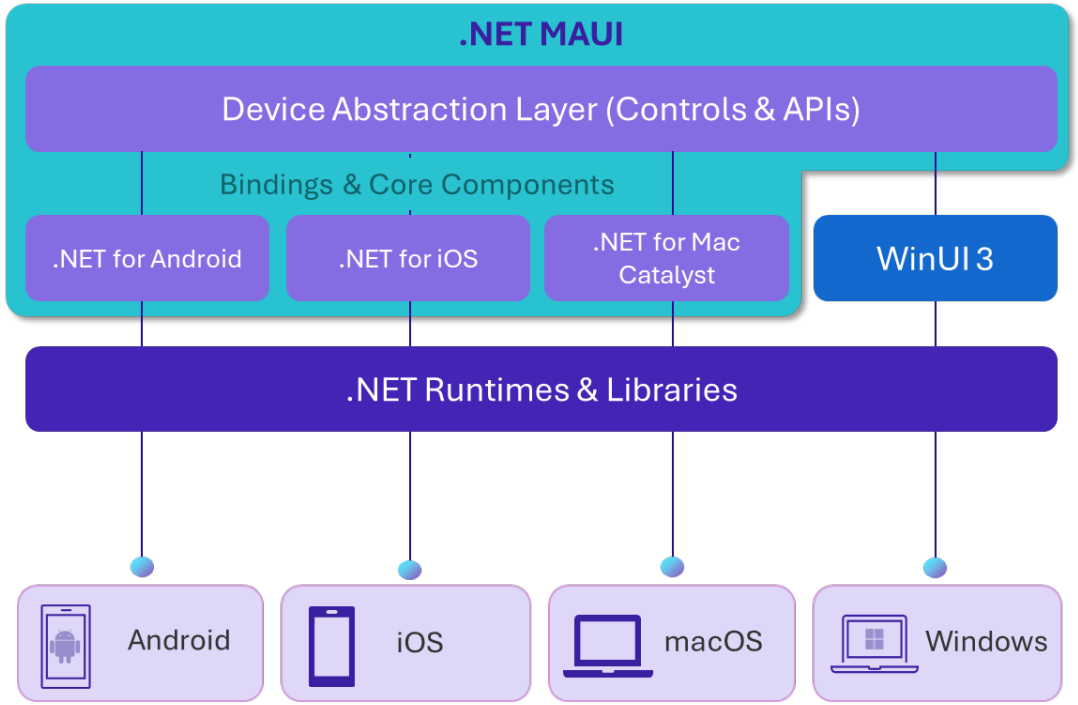


github.com/**dotnet/maui**

.NET Multi-platform App UI (MAUI)

Build native, multi-platform apps all in one framework.

- Cross-platform native mobile and desktop apps for Android, iOS, Mac, Windows
- Modern native client stack for .NET
- Single project, single codebase
- Native controls and styling
- Access over 60 common device APIs
- Access platform specific APIs if needed
- Built on top of .NET, with access to Base Class Libraries
- App store distribution and reach
- Productivity tools for Visual Studio and VS Code
- Platform is open source on GitHub
- Healthy ecosystem of controls and components



What are hybrid apps and why build them?

01 **Blend of native and web technology:** UI is written using web technologies such as HTML, CSS, and JavaScript wrapped in a native app container.

02 **Access to native platform features and device hardware:** leverage native platform features and device hardware that a web application cannot access easily or even at all.

03 **Hybrid can mean mobile and/or desktop:** run on multiple device platforms without the need for completely separate development.

04 **Reach via app store distribution:** hybrid apps are native apps that can be downloaded and installed through the app stores.

05 **Code reuse across device platforms and web browsers:** achieve greater developer productivity via code reuse across devices and web browsers.

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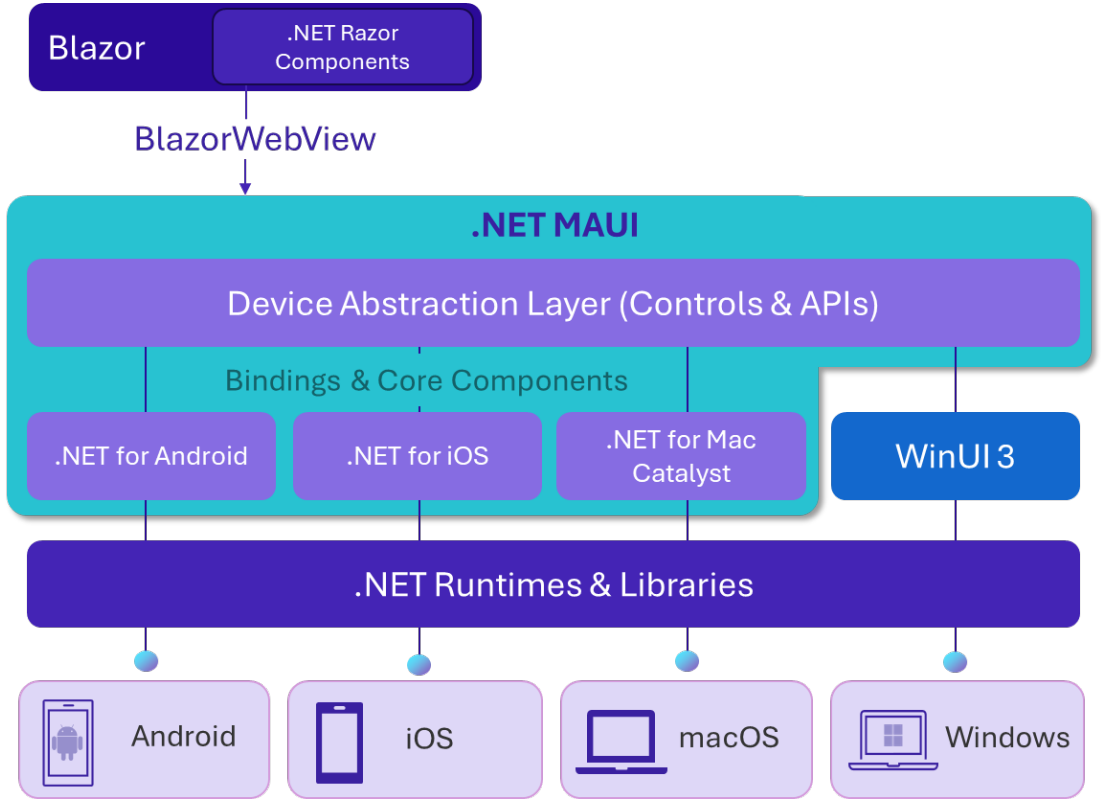
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.NET MAUI Blazor Hybrid

Build native, multi-platform apps with Blazor web UI.

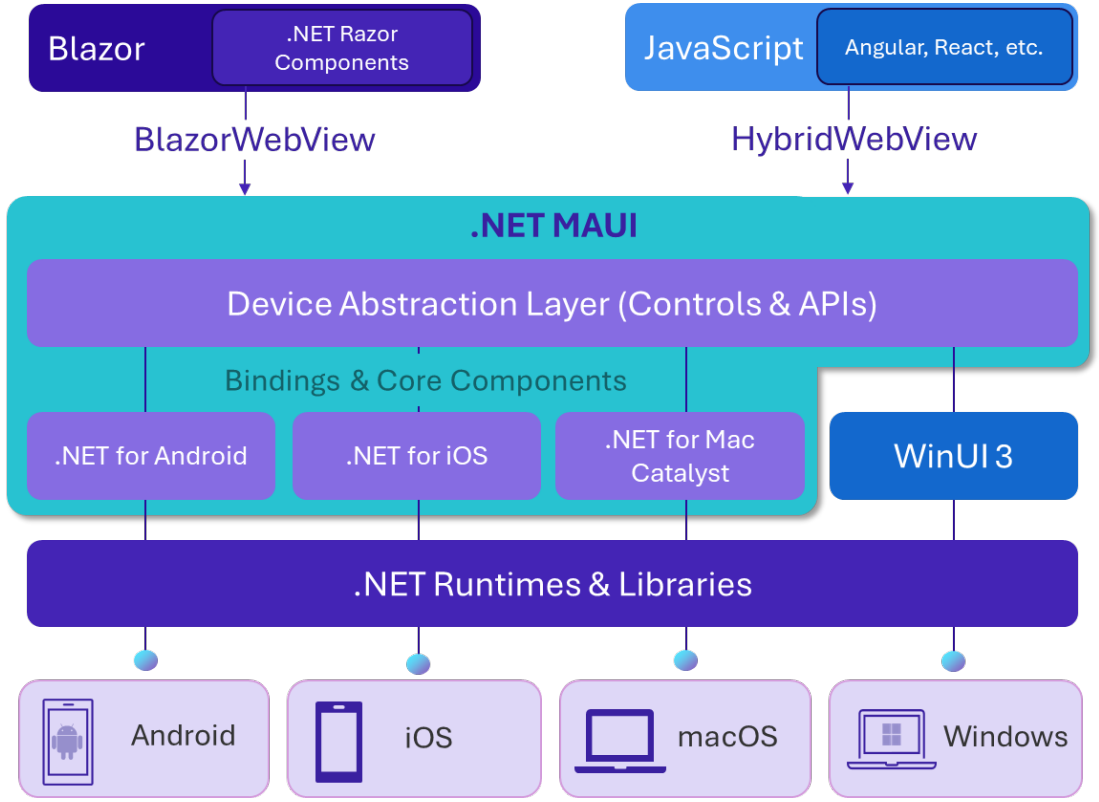
- Native app container & embedded controls
- Fully native web UI through platform WebView
- No internet required; web assets packaged with app
- Built-in Blazor input controls, validation, forms, auth
- Multi-platform with MAUI abstractions
- App store distribution and reach
- Easily reuse web UI across MAUI native and Blazor web apps with new Solution template
- … + all the benefits of .NET MAUI & Blazor productivity tooling and unified build with Visual Studio & VS Code



.NET MAUI Hybrid Apps

Build native, multi-platform apps with web UI.

- Support for other JavaScript frameworks like Angular, React, Vue, etc.
- Provides Interop between C# and JavaScript
- All the benefits of hybrid apps, i.e. WebView, assets packaged with app, store distribution



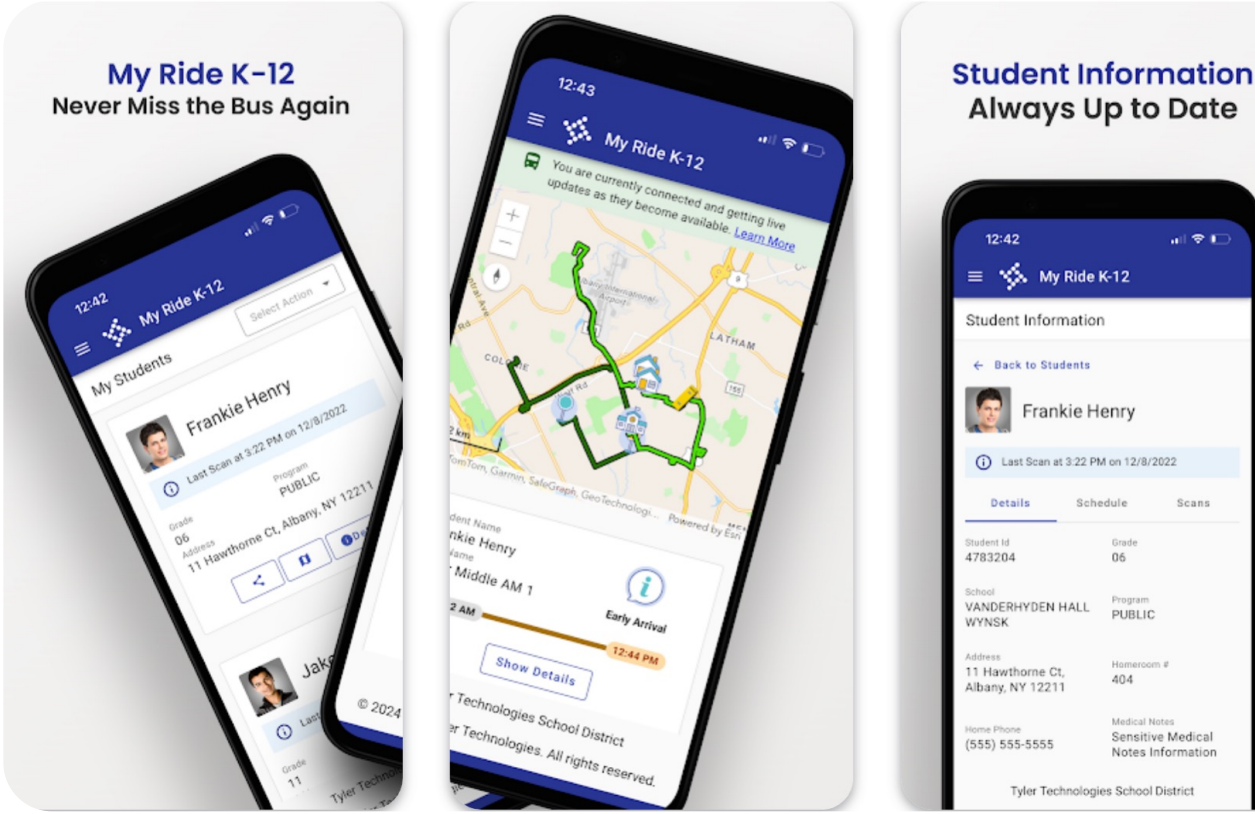
My Ride K-12

tyler technologies

.NET MAUI Blazor Hybrid & Web

> On day one, we supported web browsers, tablets, and phones. Every update goes to all platforms. For our QA people, it's just one UI to test. They verify that it works correctly on each platform, but it's the same logic and 99% the same UI across the board. The project has been an overwhelming success for us.

Chris Miller, Software Architect, Tyler Technologies, Inc.

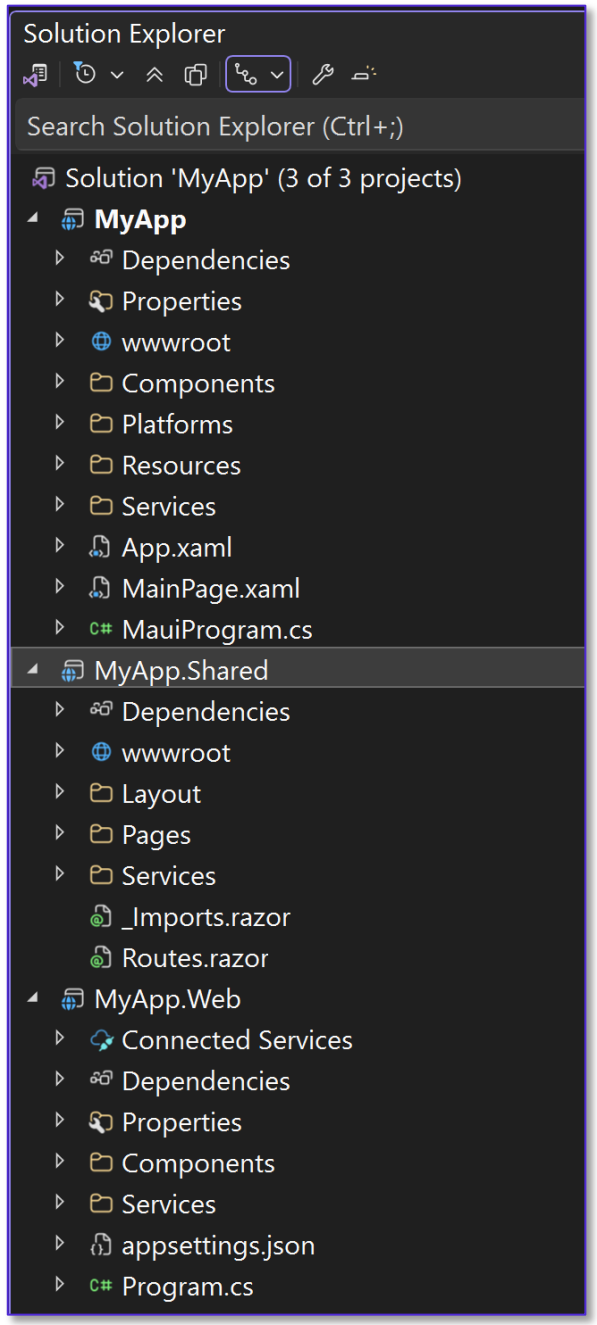


2500+
Schools

37%
Nation's students routed by app

750K
Downloads

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https://apps.apple.com/us/app/my-ride-k-12/id6443848115?platform=iphone



.NET MAUI Blazor Hybrid and Web App

> dotnet new maui-blazor-web|

Native app

MAUI hybrid app
MyApp

Shared UI & Interfaces

Razor class library
MyApp.Shared

Web app

Blazor web app
MyApp.Web

- Reuse your UI across MAUI native and Blazor web apps
- Use interface injection to support different hosting models
- Use multi-targeting to create UI for different platforms

MAUI hybrid apps in .NET 10

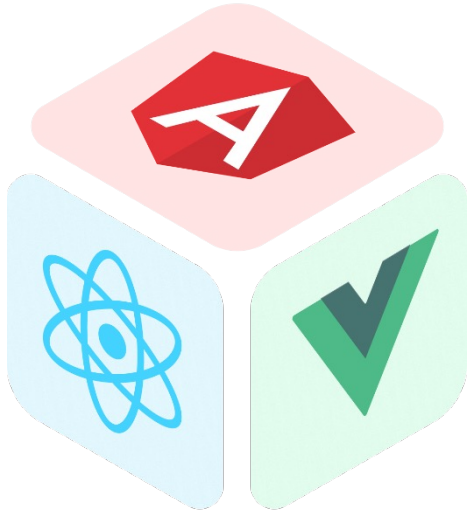
Hybrid apps with Blazor

- Updating solution template for Blazor Hybrid+Web
- Tighter inner loop with Razor and Hot Reload improvements
- More control over the BlazorWebview



Hybrid apps with JavaScript frontends

- Fit-and-finish work, packaging hybridwebview.js
- Better exception handling
- New hooks for intercepting web requests



DEMO

.NET MAUI Blazor Hybrid *and Web*



Summary

- Hybrid apps are great for web devs that want native platform capabilities & app store distribution
- Reuse web UI across your web apps and native apps with component libraries and interfaces
- Use MAUI for multi-platform abstractions and multi-targeting
- Go beyond forms over data - think about how you can leverage intelligence in your apps

Resources

Docs:

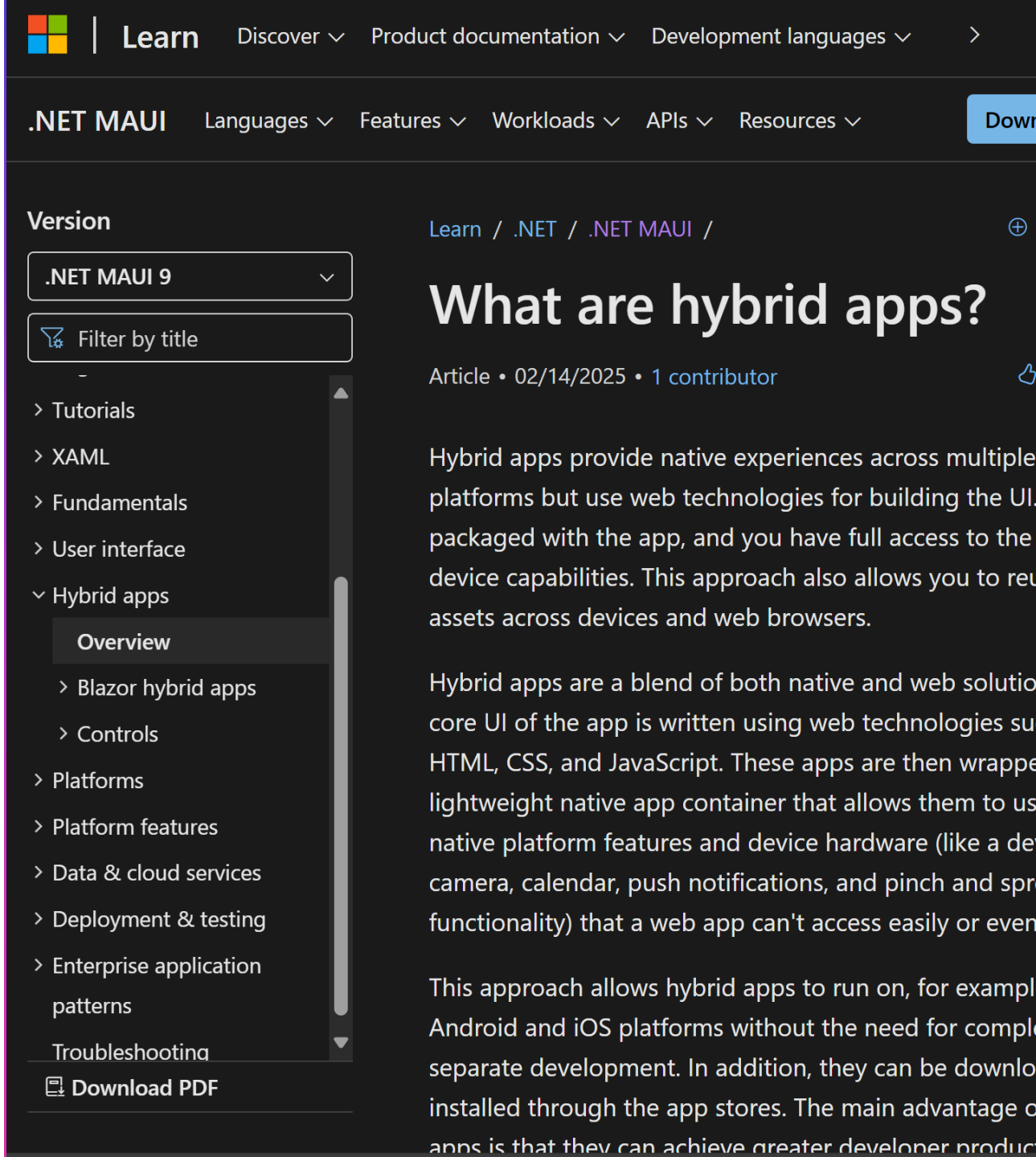
learn.microsoft.com/dotnet/maui/hybrid-apps

Repo:

github.com/dotnet/maui/labels/area-blazor

Lab:

https://github.com/dotnet-presentations/build-2025-lab305



Questions?

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